

POOJA KIRAN

SECURITY ENGINEER | AI SECURITY | SECURITY ARCHITECTURE

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Professional Summary

Security Engineer with 2+ years of hands-on experience in **AI security**, **detection engineering**, and **cloud security**. Focused on designing **security controls** and **validation frameworks** for **agentic AI**, **MCP toolchains**, **AWS IAM**, **LLM** applications, and model supply-chain risks, with emphasis on **secure system design**, **access control**, data-flow constraints, and **adversarial testing**. Developed and validated **deterministic security harnesses** including **50+ MCP detection rules**, **25 AWS IAM security rules**, and hundreds of automated security tests.

Skills

Security Architecture & Controls: Secure system design, security architecture, security controls, threat modeling, systems design, access control, authorization, least privilege, policy enforcement, runtime security, network-egress controls, data exfiltration detection

AI Security: Agentic AI, AI agent security, MCP security, prompt-injection detection, LLM security evaluation, model supply-chain security, data-poisoning detection, adversarial testing, validation frameworks

Detection & Observability: Detection engineering, SIEM, Elastic Security, ECS, security telemetry, audit logging, attack simulation, monitoring

Cloud & Infrastructure: AWS IAM, Kubernetes, Docker, Terraform, GitHub Actions, Elasticsearch, Kibana, Filebeat

Languages, Frameworks & Tooling: Python, FastAPI, scikit-learn, pytest, Hypothesis, CodeQL, SARIF 2.1.0, JSON Schema, Bandit, Trivy, pip-audit, Dependabot, Ruff

PROFESSIONAL EXPERIENCE

Independent AI Security Researcher & Engineer | Self-Directed Research

Aug 2024 - Present

- Architected and developed **AI security tooling** across **MCP**, **AWS IAM**, and **LLM red teaming**, translating attack paths into **deterministic security controls** and **automated validation workflows**.
- Engineered **50+ MCP prompt-injection rules** and **25 AWS IAM security rules** covering **AI-agent tool-call** and **cloud-identity** attack surfaces.
- Validated security tooling with **622 MCP tests** and **168 LLM security tests** through **automated regression testing** and **coverage-driven validation**.

Business & Compliance Lead | AEROSSEC | Technology Innovation Lab Externship

Aug 2025 - Dec 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ, USA

- Led **business and compliance strategy** for AEROSSEC, focusing on **third-party system security**, **compliance**, and commercialization readiness.
- Developed a funding model that secured **\$120K** in year-one commercialization funding.
- Created a **5-year financial projection** modeling growth, costs, and profitability to support commercialization planning.

Arizona State University

Jan 2025 - Oct 2025

Graduate Teaching Assistant - IT Grader

Tempe, AZ, USA

- Evaluated **85 undergraduate web-development submissions** per semester against technical requirements, **security concepts**, and implementation quality.
- Assessed **52 graduate-level IT-security assignments** using rubric-based evaluation of **security policies** and **cybersecurity concepts**.
- Collaborated with faculty to provide technical feedback supporting students' **secure coding practices**.

PROJECTS

MCP Agent Security Gateway [GitHub](#)

Oct 2025 - Sep 2026

• **Python**, **MCP**, **JSON-RPC 2.0**, **FastAPI**, **Docker**, **Kubernetes**, **Elastic/SIEM**

- Designed an inline MCP security gateway inspecting **tools/call** traffic between AI agents and downstream MCP servers, enforcing layered policies across server trust, tool calls, process-spawn intent, semantic intent, and network egress.
- Engineered 50+ prompt-injection detection patterns with Unicode/encoded-input normalization, capability checks, PII/exfiltration signals, tool-output inspection, SHA-256 hash-chained audit logging, rate limiting, circuit breakers, and security telemetry.
- Validated 622 automated tests at 78% coverage; integrated 9 Elastic detection rules, 6 Atomic Red Team-style attack simulations, and 21 SIEM tests for detection and security-event validation.

AWS Agent Identity Guard [GitHub](#)

Apr 2025 - Sep 2025

• **Python**, **AWS IAM**, **SARIF 2.1.0**, **GitHub Code Scanning**, **Hypothesis**

- Built a static analyzer for AI-agent IAM policies with 25 deterministic security rules covering wildcard permissions, iam:PassRole, trust policies, resource scoping, Bedrock/SageMaker permissions, audit-trail tampering, and permission boundaries.
- Implemented text, JSON, and SARIF 2.1.0 findings with GitHub Code Scanning integration and CI policy gates for high- and critical-severity IAM violations.
- Verified all 25 rule IDs through positive/negative tests, parser edge cases, Hypothesis fuzz testing, failure-mode tests, and SARIF validation; recorded 230 passed tests.

HF Model Provenance Scanner [GitHub](#)

Nov 2024 - Mar 2025

- **Python**, Pickle, SafeTensors, GGUF, ONNX, MITRE ATT&CK; v19
- Built a static model-supply-chain scanner for Hugging Face repositories covering provenance, impersonation, pickle risk, and supply-chain indicators without executing model artifacts.
- Implemented analysis across model and configuration artifacts for risks including pickle, SafeTensors, GGUF, ONNX, Keras, typosquatting, dependency confusion, token exposure, and malicious model repositories; mapped findings to MITRE ATT&CK; v19.
- Validated detection against 33 committed red-team fixtures spanning 12 documented incident reproductions, 18 extended attack variants, and 3 large-scale fixtures, achieving 33/33 detection with reproducible fixture-level evidence.

LLM Red-Team Framework [GitHub](#)

Jul 2024 - Oct 2024

- **Python**, TF-IDF, Logistic Regression, FastAPI, SARIF, GitHub Actions
- Built an offline adversarial evaluation framework spanning 6 prompt-injection attack categories, converting attack scenarios into repeatable evaluation datasets and regression tests.
- Implemented TF-IDF + Logistic Regression with grouped template splitting to reduce train/test leakage, plus structured SARIF reporting for CI security validation.
- Validated detector behavior with 168 tests and 96% coverage; achieved F1=0.97 on grouped held-out templates and F1=0.83 on novel-phrasing out-of-distribution evaluation, quantifying the generalization gap.

ML Security Benchmark Suite [GitHub](#)

Jan 2024 - Mar 2024

- **Python**, Pytest, JSON Schema, GitHub Actions
- Built a cross-project ML security regression harness using typed adapters, versioned fixtures, declarative contracts, and shared JSON Schema to detect interface and behavioral regressions across security tools.
- Integrated 4 security-tool adapters spanning IAM analysis, model scanning, prompt-injection detection, and dataset-poisoning analysis with structured results and automated schema validation.
- Validated the suite through 67 test functions across 8 test modules; an HF-scanner run over 3 known-bad and 2 known-good committed fixtures achieved precision=1.0, recall=1.0, and F1=1.0, scoped strictly to that fixture set.

EDUCATION

Arizona State University
M.S., Information Technology
• GPA: 3.87/4.00

Aug 2024 - May 2026

Tempe, AZ, USA

Ramaiah University of Applied Sciences
B.E., Computer Science & Engineering
• GPA: 8.44/10

Aug 2019 - Aug 2023

Bangalore, KA, IN

CERTIFICATIONS

- AWS Academy Graduate - **Cloud Security Foundations** (2025)
- AWS Academy Graduate - **Cloud Architecting** (2025)

PUBLICATIONS

- **A Personalized E-Learning System Using Reinforcement Learning Through Satellite.** IEEE INDICON, 2023/2024 | Document ID: 10440852